Fabrication and Spectroscopic Analysis of Solid State Tesla Coils

J. Raugh, J. Scholl, and A. Thoms

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Abstract

In this experiment, we fabricated a solid-state Tesla coil and spectroscopically analyzed the sparks emitted from its topload. We successfully constructed a Tesla coil and recorded the sparks discharged under two conditions: with an uncontaminated topload, a nail without any coating, and with a salt-coated topload, a nail that was soaked in an aqueous salt solution. A device known as the RSpecExplorer analyzed the sparks using a webcam with a built-in diffraction grating and computer software with video recording and spectroscopic analysis features. The RSpecExplorer observed that the uncontaminated topload had a spectrum with violet color ranging from 385.88 to 396.26 nm, blue from 418.34 to 427.9 nm, and orange from 593.54 to 595.24 nm. In the salt-coated spectrum, a violet color ranging from 395.62 to 396.16 nm was observed, blue from 424.56 to 428.54 nm, yellow from 395.62 to 396.16 nm, and red from 597.64 to 606.9 nm.

1 Introduction

The objective of the experiment is to construct a Tesla coil and record the spectroscopy of the spark produced. Construction of the Tesla coil consists of electrical and physical components.

The electrical components are in the circuit, wired with diodes, resistors, capacitors, and a MOSFET. The primary components are the primary and secondary coils, ballast, and the topload where the spark is discharged. A spectrum can be observed when the spark interacts with air and is coated in a salt solution. Data is collected with the RSpec Explorer, which can identify peaks in a spectrum using a camera with a built-in diffraction grating. The observed peaks can be compared to a known spectrum, such as O_2 and N_2 for ionized air and sodium for the aqueous salt solution.

1.1 History

Tesla coils are electrical resonant transformer circuits that can generate high-voltage, low-current, high-frequency alternating-current electricity [1]. Nikola Tesla invented and patented the Tesla coil in 1891, expanding on the first experiments of electromagnetism from Heinrich Hertz. Tesla visualized his work as a potential contributor to bigger, more wild inventions such as a death ray; ultimately, he aimed to achieve wireless power [2]. In the last hundred years, the role of Tesla coils as a power supply has become obsolete due to cheaper and simpler alternatives. Examples of modern uses for the Tesla coil are electrotherapy or similar psuedomedical practices, educational displays, music or entertainment, etc [1]).

The other major component of this experiment is spectroscopy. In 1666, Isaac Newton demonstrated dispersion, a phenomenon where sunlight or white light is split into a continuous, colorful spectrum by a prism [5]. This spurred other scientists to further investigate the behavior of light and conduct similar experiments. Nearly 150 years later, William Herschel discovered the infrared spectrum by observing how wavelengths of light change based on exposure to temperature [4]. There were many other significant contributions to spectroscopy in the following years, such as Young's double slit experiment, the discovery of the ultraviolet spectrum, etc. In 1814, Joseph Fraunhofer observed dark lines in the solar spectrum. These dark lines, now acknowledged as "Fraunhofer lines", are absorption lines in the continuous spectrum of the sun's blackbody radiation [7]. Gustav Kirchoff and Robert Bunsen delved

further into Fraunhofer's findings and in 1859, realized that spectroscopy can serve as an identification tool in chemical analysis.

2 Theory

2.1 Tesla Coils

The principle of electromagnetic induction is responsible for Tesla coil's ability to generate high voltages on low currents. Alongside these high voltages, an electric field is produced and resonance can be achieved [6]. Resonance occurs when the primary and secondary coils are tuned to transfer current simultaneously with the moment of maximum energy [1]. The equation that represents the condition for resonance is

$$L_1 C_1 = L_2 C_2 \tag{1}$$

where L is the magnetic inductance in Henries (H), C is the capacitance in Farads (F), and 1 and 2 are the first and second coils. To measure L and C, it is dictated by equations

$$L = \frac{r^2 N^2}{9r + 10l} \tag{2}$$

where L is inductance in μH, r is the radius in inches, N is the number of turns, and l is the length; and,

$$C = b \cdot 2 \cdot r \tag{3}$$

where C is capacitance in pF, b is a constant, and r is the radius in centimeters [16]. Due to the oscillatory nature of the energy from the primary coil, the secondary coil induces a current, and a magnetic flux is emitted between the coils. The specific frequency at which the primary and secondary coils are tuned to oscillate allows for maximum energy transfer

between them is known as the resonance frequency. The resonance frequency of a Tesla coil can be calculated with the equation

$$f = \frac{1}{2\pi\sqrt{LC}} \tag{4}$$

where f is the frequency in Hertz (Hz), L is the inductance in Henries (H), and C is the capacitance in Farads (F). The emission of voltage occurs as sparks, or the discharge of light when the electric field strength overcomes the dielectric strength of air [1]. Voltage can be measured with the equation

$$V_2 = V_1\sqrt{C_1 \cdot C_2} = V_1\sqrt{L_2 \cdot L_1} \tag{5}$$

where V is the voltage in volts (V), L is the magnetic inductance in Henries (H), C is the capacitance in Farads (F), and 1 and 2 are the first and second coils. Tesla coils are typically built with a power supply, a spark gap, a capacitor, a toroid (hollow circular ring), and primary and secondary coils. However, in our experiment, we won't use a spark gap; instead, we'll utilize a MOSFET to control the circuit.

2.2 Spectroscopy

Spectroscopy is the study of the interaction between light and materials. Due to the properties of a material, each element possesses a characteristic spectrum due to its unique emission and absorption behaviors. No two spectra, whether it be the absorption or emission lines of two different elements, are identical. The characteristic nature is attributable to the unique composition of subatomic particles within each atom of an element. This unique combination of spectral lines that are emitted from an element is called the emission spectrum. When the atoms of an element become excited when exposed to heat or electricity, the emission spectrum can be observed. When an atom is excited, it releases energy in the form of photons. For instance, an excited electron with lower energy would emit a photon of longer wave-

length, such as red light and vice versa would result in a photon of shorter wavelength, like violet light. A diffraction grating can split the emitted light into several, symmetrical orders of emission spectra. These orders are caused by the transmission of light bending through the slits of a diffraction grating, resulting in a pattern of spectral lines from constructive wave interference.

The color of the spark is due to a phenomenon known as ionized-air glow. A discharge of high voltage from the toploaded causes the surrounding air to be ionized. Atoms in the air gain kinetic energy, caused by excitement from the electric field [11]. When atoms collide, electrons can become dislodged, leaving the atoms in an ionized state. The ionized atoms continue to randomly collide but, will collect a replacement electron [8]. When an atom recombines with an electron, neutralization occurs, often resulting in the emission of a spark if the electron cascades down energy levels and emits a photon [11]. We expect to observe the spectral lines characteristic to O_2 and N_2, which are the most prominent elements in air.

3 Procedure

3.1 Construction of the Tesla Coil

The Tesla coil was constructed from several electrical and physical components, listed in Table 1.

Table 1: Components of Tesla Coil

Amount	Component
1	MOSFET with heat sink
1	Power diode
1	12V bidirectional TVS diode
1	400V bidirectional TVS diode
1	470pF capacitor
1	Ballast, lightbulb* and toaster**
1	1k, 1/4W resistor
2	15k, 1/4W resistor
2	Coils, primary and secondary
1	Nail, topload
1	Pegboard

* Multiple light bulbs of various wattages were tested.
** Not included in the final design.

The physical components are the coils, ballast, topload, and pegboard. Each coil had a specific number of turns: the primary coil had 2.5 turns, while the secondary coil had 227 turns. At the top end of the primary coil is the topload. A nail was chosen to serve as the topload, since its pointed end is advantageous for initiating sparks. The ballast regulates the amount of current. Various components were tested to determine the ideal ballast. Table 2 records ballasts of different wattages and their observed effects on the topload.

Table 2: Tesla Coil Spark Size [15]

Ballast Power (W)	Total Circuit Current (A)	Spark Quality
14	N/A	No Spark, Flickering Bulb
15	N/A	No Spark, Steady Bulb
26	N/A	No Spark, Steady Bulb
40	2.3	Very Small Spark ($\sim \frac{1}{8}$ in), Not Stable, Bright Bulb
75	1.9	$\sim \frac{1}{4}$ in Spark, Very Stable, Bright Bulb
100	1.33	$\sim \frac{1}{4}$ in Spark, Brighter, Stable with Grounding, Bright Bulb
700W (Toaster)	N/A	Large, Unstable Sparks, Blew the Circuit after 30s

Incandescent light bulbs between 14 and 100 Watts and a toaster of 700 Watts were the main subjects of testing. As described in Table 2, the ballasts between 14 and 26

watts did not generate a spark while the toaster produced too many, resulting in a blown circuit. The preferred ballasts were lightbulbs in the range of 40-100 Watts, producing sparks that extended approximately 1/8-1/4 inches. The physical components were arranged on a pegboard, alongside the electrical components (circuit).

Figure 1 depicts the circuit used as the model for our Tesla coil.

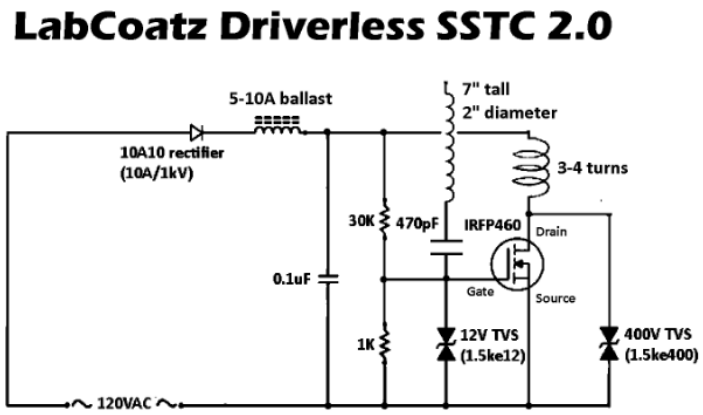


Figure 1: The model for the Tesla coil circuit [10].

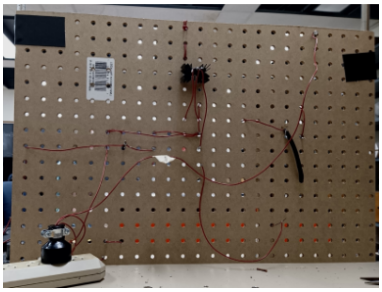
The MOSFET controls the circuit by handling the high power levels. An ideal MOSFET for this circuit should be able to withstand continuous currents over 5A, peak currents over 20A, and can dissipate over 250W of power [10]. To prevent the MOSFET from over-voltage or voltage spikes, multiple diodes were included in the circuit. The resistor array was comprised of one 1k and two 15k resistors, totaling 31k Ohms. A resistor array over 50k Ohms may cause the MOSFET to fail [10].

Figure 3 illustrates the stages of completion of the Tesla coil circuit.

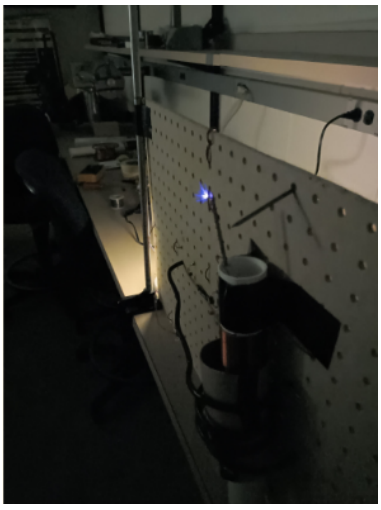
Prior to obtaining all the materials, the circuit was planned out using labels and the available components. Construction of the Tesla coil commenced once all the necessary components were gathered. The wires and the electrical components were weaved through the pegboard for organization and stability, and connected together using a soldering iron. Any exposed wire or soldered parts were wrapped in electrical tape. A grounded nail was added to assist in managing the sparks emitted by the top load, while a separate top load



(a) The planning phase of the circuit.



(b) The backside of the pegboard, featuring the main components of the circuit



(c) The Tesla coil in operation. Discharges of sparks are emitted from the topload.

Figure 2: The stages of construction for the Tesla coil circuit [15].

nail, coated in an aqueous salt solution, was prepared as an alternative to the standard one. The power source for the Tesla coil system is the 120V wall outlet.

3.2 Spectroscopy of the Tesla Coil

The apparatus for the spectroscopy portion of the experiment was the Rspec Explorer equipment, as shown in Figure 3.

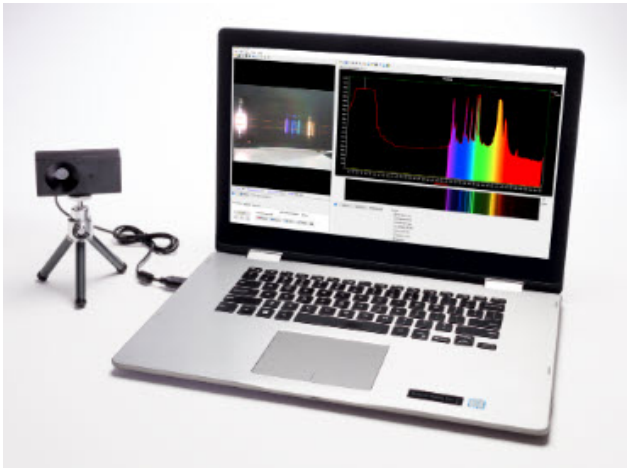


Figure 3: The Rspec Explorer camera and the computer application [13].

The RSpec Explorer camera has a built-in diffraction grating that allows the diffracted light to be observed in the computer application. A demonstration of the RSpec Explorer computer program and its features are displayed in Figure 4.

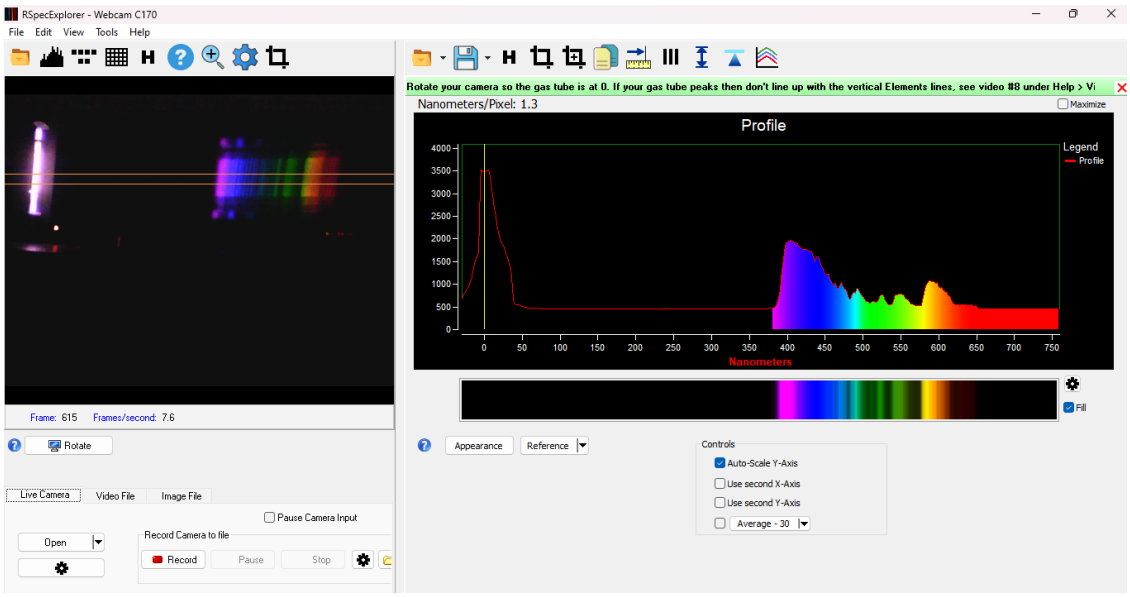


Figure 4: The RSpec Explorer computer utilizing the webcam to diffract an air gas-discharge lamp. The spectrum on the right side of the screen was derived from the webcam.

In Figure 4, a spectrum is graphed from the wavelengths in nanometers and the intensity in pixel counts. The spectrum was created from the light diffracted from the light source. To control the amount and intensity of light analyzed by the program, the distance between the yellow horizontal lines in the camera's field of view can be adjusted. This can alter the precision of the spectrum reading. For accuracy, the red line, or intensity peak, was aligned with the yellow line at 0 nanometers as closely and consistently as possible for each trial. Each trial consisted of a video recording roughly 5-10 seconds long of the diffracted light. Both the uncontaminated air-ionizing topload and the salt-covered topload underwent 5 trials each. From the video recordings, measurements of the observed peaks were identified from the individual frames. The most prominent peaks had 5 data points recorded for each trial, while the relatively subtle peaks were recorded on a case-by-case basis.

4 Analysis and Results

4.1 Analysis

Peaks for the uncontaminated topload and the sodium-coated topload were recorded for 5 frames of each trial. In some of these frames, the camera was not fully calibrated, meaning the intensity peak was not aligned at zero. This problem was counteracted by subtracting or adding the offset to the wavelength values. The calibrated wavelengths were averaged based on the peaks, and a range of values was determined. All analysis was executed in `Microsoft Excel`. No error was calculated for the experimental values since there were no exact expected values for comparison. However, the observed spectrum was compared to the known spectrum of suspected elements. For a broad comparison, the wavelength ranges for each peak were compared to known values rather than the calculated averages. The elements that were suspected to be present were O_2 and N_2 for the uncontaminated, air-ionizing topload and sodium for the salt-coated topload.

4.2 Results

An example of the overarching spectral lines for the uncontaminated topload is presented in the spectrum in Figure 5.

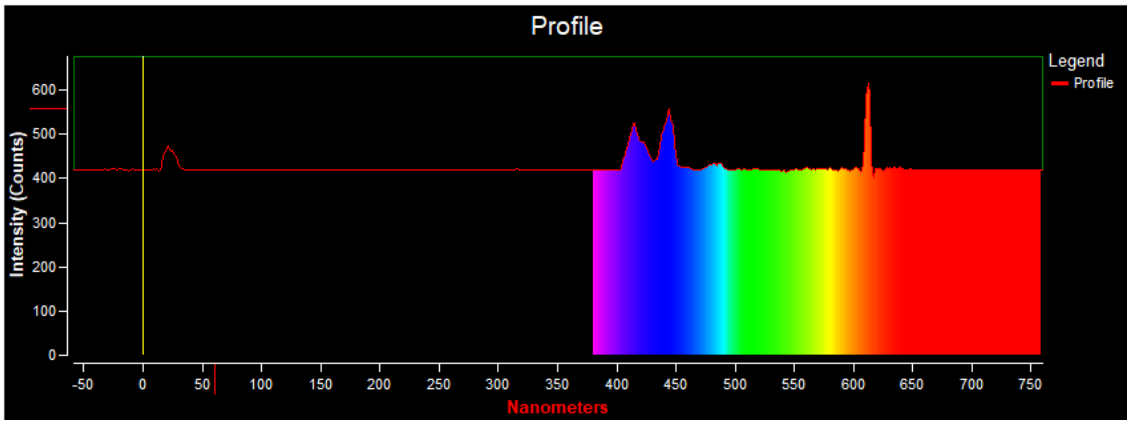


Figure 5: The graph depicts the emission spectrum from the sparks produced by the uncontaminated topload. The image is generated by the RSpec Explorer program and represents a frame captured during one of the trials.

In Figure 5, it is evident there are prominent peaks of three colors: violet, blue, and orange. From each trial, the wavelengths of observed peaks were recorded for 5 different frames. The measured wavelengths were calibrated and an average value per color was calculated, in addition to a range of values. An average and a range of values were determined for the violet, blue, and orange colors, as seen in Table 3.

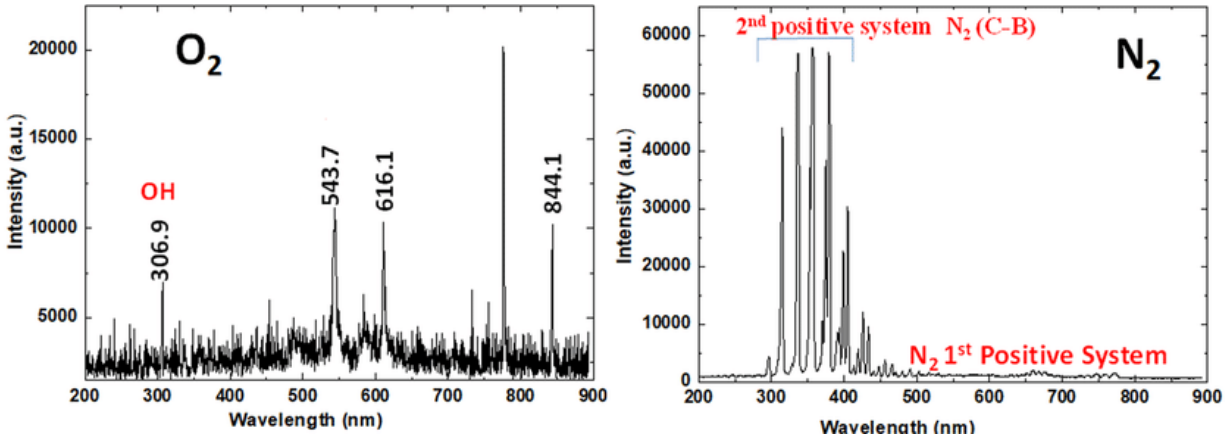
Table 3: Observed wavelengths for ionized air

Color	Average (nm)	Range (nm)
Violet	392.86	385.88-396.26
Blue	424.84	418.34-427.9
Yellow-Orange	594.35	593.54-595.24

The violet color has an average wavelength of 392.86 nm and a range of 385.88 to 396.26 nm, the blue has an average of 424.84 nm and a range of 418.34-427.9 nm, and the orange has an average of 594.35 nm and a range of 593.54 to 595.24 nm. The average and range of the calibrated wavelengths can be compared to the known or expected values of O_2 and N_2. In Figure 6, the spectral peaks that were observed through the RSpec Explorer camera are compared to a known spectrum of O_2 and N_2 from photoionized plasma.



(a) The observed spectral lines of ionized air from an RSpec Explorer frame.



(b) A known spectrum of O_2 and N_2 from photoionized plasma [**?**].

Figure 6: The observed spectral lines of ionized air vs. a known spectrum of photoionized O_2 and N_2.

Since the known spectra of O_2 and N_2 in Figure 6b were observed under different conditions than in this experiment, the comparison may be imprecise due to these discrepancies. The purpose of the comparison is not to pinpoint exact values but to determine if the suspected elements are present. A comparison between the known wavelengths drawn from the O_2 and N_2 spectra in Figure 6 and the calibrated wavelengths measured by the RSpec Explorer is displayed in Table 4.

Table 4: Ionized air, expected vs. experimental

Color	Expected (nm)	Experimental (nm)
Blue	<450	424.56-428.54
Violet	>300	395.62-396.16
Red-orange	543.7-616.1	593.54-595.24

The RSpec Explorer camera is only capable of capturing wavelengths in the visible light spectrum between 350 and 400 near the ultraviolet side and 750 near the infrared. Most of the N_2 spectrum falls within the ultraviolet range; however, it still emits wavelengths in the visible light spectrum as violet and blue. The range of the blue wavelength observed is

less than the maximum peak of blue (<450 nm) in the known N_2 spectrum. Similarly, the range of the violet wavelength observed exceeds the minimum peak of violet (>300 nm) in the known N_2 spectrum. It can be determined that the violet and blue peaks observed in the experimental data resemble those of the N_2 spectrum. The known O_2 spectrum from Figure 6 has several peaks, mainly in the orange to infrared range. As discerned from Table 4, the experimental range of values falls between the two red-orange peaks, ranging from 543.7 to 616.1 nm. It can be determined that the orange peaks observed in the experimental data resemble those of the O_2 spectrum.

An example of the reoccurring spectral lines of the salt-covered topload is presented in the spectrum in Figure 7.

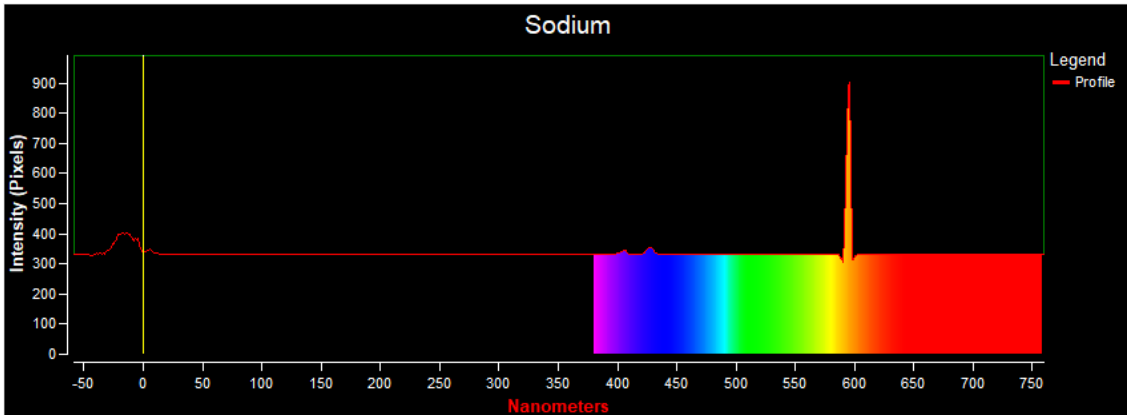


Figure 7: The graph depicts the emission spectrum from the sparks produced by the salt-coated topload. The image is generated by the RSpec Explorer program and represents a frame captured during one of the trials.

Prominent peaks of three colors can be identified in Figure 7: a sharp yellow line and two weaker peaks of blue, violet, and occasionally red. If an offset was present in the frame, the raw wavelengths were calibrated to compensate for it. An average value per wavelength and a range of wavelength values were calculated with the calibrated data. These average and range values for the violet, blue, yellow, and red colors are shown in Table 5.

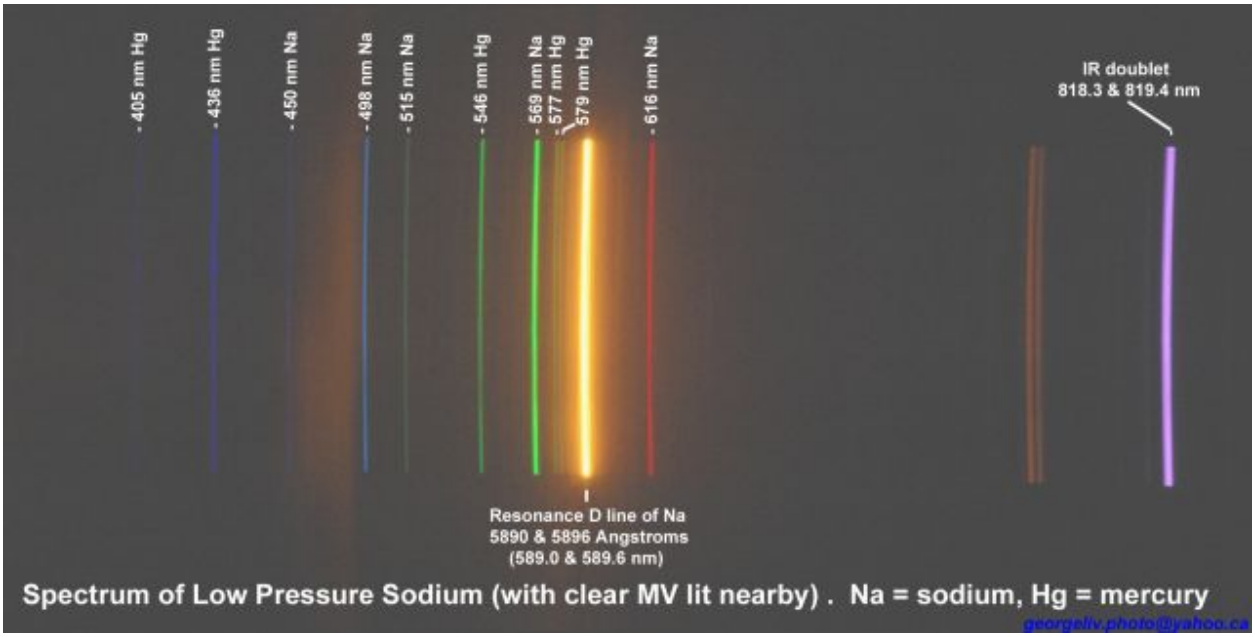
Table 5: Observed wavelengths for sodium

Color	Average (nm)	Range (nm)
Violet	395.89	395.62-396.16
Blue	426.55	424.56-428.54
Yellow	588.9	575.74-596.02
Red	602.2	597.64-606.9

The violet color has an average wavelength of 395.89 nm with a range of 395.62 to 396.16 nm, and the blue has an average of 426.55 nm with a range of 424.56 to 428.54 nm. The yellow color has an average wavelength of 588.9 nm, with a range of 395.62 to 396.16 nm, while the red color has an average wavelength of 602.2 nm, with a range of 597.64 to 606.9 nm. Errors were not calculated due to the measured lines lacking the exactness necessary to be confidently ascertained. Instead, a range of values is preferred for comparison purposes. In Figure 8, the observed spectral peaks from the RSpec Explorer apparatus are compared to a known spectrum of sodium from a sodium-vapor lamp. In both spectra, a bright yellow line is visible.



(a) The observed spectral lines of the salt-coated nail from an RSpec Explorer frame.



(b) A known spectrum of sodium from a sodium-vapor lamp.

Figure 8: The observed spectral lines of the salt-coated nail vs. a known spectrum of sodium from a sodium-vapor lamp [14]

The known spectra of sodium in Figure 8b were observed under different circumstances than in this experiment, meaning any drawn comparison is aimed at determining similarities rather than precise wavelength identification. A comparison between the known wavelengths drawn from the sodium spectrum in Figure 8 and the wavelengths measured by the RSpec Explorer is shown in Table 6.

Table 6: Sodium, expected vs. experimental

Color	Expected (nm)	Experimental (nm)
Yellow/Doublet	589.0/589.6	575.74-596.02
Red	616	597.64-606.9
Blue	<450	424.56-428.54
Violet	>300	395.62-396.16

The sodium spectrum mainly consists of a bright yellow doublet and a relatively faint

red and green line. The yellow doublet has expected values of 589.0/589.6 nm, whereas the experimental data from RSpec Explorer measure a range of 575.74-596.02 nm, drawn from Table 5. Since the expected value and the experimental range are comparable it can be determined that the yellow line from the experimental data resembles the yellow doublet in the sodium spectrum. A red wavelength, which is also characteristic of the sodium spectrum, was occasionally observed in some of the trial frames. According to Table 5, the observed range of the red wavelength (597.64 to 606.9 nm) does not align with the expected wavelength of 616 nm. It is possible that the observed red color could be the red wavelength of sodium; however, due to the inconsistency, there are alternate explanations such as an anomaly consequence of a mishap during the data collection process. In Table 5, there is an additional blue and violet line. The violet and blue wavelengths resemble the values observed in the previous data set. This suggests that in addition to sodium emission, the spark is ionizing the air simultaneously, resulting in observable spectral lines presumably from air. A resemblance indicates that the violet and blue peaks observed in the experimental data may be characteristic of those from the N_2 spectrum.

5 Conclusion

The objective of this lab was to build a Tesla coil and to analyze the sparks it produced via spectroscopy. Fabricating the Tesla coil required an understanding of its fundamental physics and the uses of the electrical components necessary to build its circuit. Additionally, a basic understanding of spectroscopy concepts was applied for measuring and analyzing the quantitative data. The RSpec Explorer equipment, consisting of a camera with a built-in diffraction grating and a computer program, was utilized for data collection. Using the RSpec Explorer meant familiarization with the equipment and the ability to execute its features, such as video recording with the webcam and adjusting the parameters for analysis. The spectrum is graphed in nanometers per pixel.

The RSpec Explorer apparatus observed peaks of violet, blue, and orange in the spectrum emitted by the sparks produced from the uncontaminated topload. The violet wavelength has an average of 392.86 nm with a range of 385.88 to 396.26 nm, the blue has an average of 424.84 nm with a range of 418.34-427.9 nm, and the orange has an average of 594.35 nm with a range of 593.54 to 595.24 nm. It is determined that the observed wavelengths resemble spectral lines characteristic of a known O_2 and N_2 spectrum of photoionized plasma. The violet and blue color is attributed to the known N_2 spectrum, while the orange color is similar to the spectral lines in O_2.

For the sparks discharged by the salt-coated topload, the RSpec Explorer observed peaks of violet, blue, yellow, and red in the spectrum. The violet color has an average wavelength of 395.89 nm with a range of 395.62 to 396.16 nm, and the blue has an average of 426.55 nm with a range of 424.56 to 428.54 nm. The yellow color has an average wavelength of 588.9 nm, with a range of 395.62 to 396.16 nm, while the red color has an average wavelength of 602.2 nm, with a range of 597.64 to 606.9 nm. It is determined that the observed wavelengths resemble spectral lines characteristic of a known sodium spectrum from a sodium-vapor lamp. The yellow peak resembles the spectral lines in the sodium spectrum. The violet and blue colors are attributed to the known N_2 spectrum, indicating that the effects of air ionization are observable. A red color was observed, but its presence was too inconsistent to determine if it is potentially a result of the sodium spectrum.

6 Distribution of Work

6.1 Aubrie Thoms

- Slides 10-17 of final presentation.
- Collected data using the RSpecExplorer and analyzed the wavelengths.

6.2 Jon Scholl

- Slides 2-5 of final presentation.
- Provided theoretical knowledge about Tesla coils and circuits.

6.3 Jake Raugh

- Slides 6-9, & 17 of final presentation.
- Built the Tesla coil.

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[15] Jake Raugh, lab mate

[16] Jon Scholl, lab mate